Nihal Sirivolu

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Education

Virginia Polytechnic Institute and State University **May 2027**
B.S. in Computational Modeling & Data Analytics *Blacksburg, VA*

Experience

Edrevel AI **June 2026 – Present**
AI/ML Engineering Intern *Fairfax, VA – Hybrid*

- Engineered an **end-to-end computer vision pipeline** for workforce-training compliance platform using Python, PyTorch, OpenCV, YOLO, SAM-2, and MediaPipe to extract objects, people, hand interactions, and task evidence from video.
- Implemented **open-vocabulary detection, persistent tracking, segmentation, and keyframe extraction**, producing auditable bounding boxes, timestamps, scene assignments, and confidence metadata.
- Integrated ML inference through **FastAPI, Docker, and AWS**, connecting video uploads with backend processing, evaluator results, artifact storage, and reviewer workflows.
- **Automated Jira workflows through REST APIs** to create and update issues from model-evaluation results and test failures, attach evidence artifacts, and preserve traceability from video scene to reviewer decision.
- Designed privacy-aware AWS Rekognition person validation and maintained production data contracts, documentation, and automated validation across a suite of **600+ tests**.

IDPro AI Tourism – Virginia Tech **August 2026 – Present**
Research Team Member *Blacksburg, VA*

- Leading the team's **Meta Wearables SDK integration** for a Kotlin Android tour app, developing **Ray-Ban Meta Glasses** device registration, camera sessions, photo capture, and a phone-camera fallback for testing.
- Authoring **SDK setup and architecture documentation** covering permissions, device testing, capture contracts, and data handling so teammates can reproduce the glasses-to-backend workflow.
- Building a **Meta Project Aria ingestion pipeline** that extracts first-person tour frames, transcribes guide audio with Whisper, and links tour locations to page-cited documents.

4SightAI **June 2025 – August 2025**
Automation/Cloud Engineering Intern *San Francisco, CA – Remote*

- Built **real-time data ingestion services** for health, exercise, dietary, and business data supporting the Nutrition Co-Pilot and AI Co-Pilot Consulting platforms, **reducing processing latency by 45%** and improving model-context accuracy by **35%**.
- Created **Pandas and Power BI dashboards** for patient and operational datasets, identifying trends and producing actionable insights for wellness and consulting workflows.
- Developed **Python and Jira REST API automations** to create, categorize, assign, and update work items from ingestion failures, data-quality checks, and dashboard findings across AI product pipelines.
- Developed **AWS Lambda, S3, and DynamoDB workflows** with **vector retrieval** for unstructured-data processing, research summarization, report generation, and AI-assisted strategic recommendations.

Projects

RSNA Knee Abnormality Detection | *Python, PyTorch, DICOM, NLP, Kaggle* **August 2026**

- Developed a **multimodal learning pipeline for knee MRI abnormality classification**, integrating DICOM preprocessing, image augmentation, and multilingual radiology-report normalization in PyTorch.
- Investigated the contribution of **imaging and clinical-text features** through image-only, text-only, and multimodal baselines, using cross-validation, class-imbalance handling, and decision-threshold calibration.
- Achieved **89.43% classification accuracy** and established configuration-driven training and inference workflows to support reproducible experiments and comparative model evaluation.

Magentic.AI | *TypeScript, Node.js, MCP, LangChain, Docker* **June 2026 – Present**

- Building an **open-source MCP server** that connects AI clients to development tools and exposes 24+ tools for conversation search, context sharing, and secure handoffs.
- Implementing **LangChain model and embedding integrations** for semantic search and grounded question answering, with message-level citations and explicit responses when supporting evidence is unavailable.
- Modeling **conversations, agents, tools, and handoffs with Palantir Foundry Ontology**, while developing scope-limited agent endpoints and validating permissions, Docker builds, and model-provider fallbacks through automated tests.
- Adapting the platform into a **competition-ready agent demo** for the Virginia Tech–Accenture Build Challenge, and Microsoft for Startups, focused on secure and reusable AI context.

Technical Skills & Certifications

Languages: Python, TypeScript, Kotlin, Java, R, C/C++, JavaScript, SQL, Go
ML / Computer Vision: PyTorch, TensorFlow, OpenCV, YOLO, SAM-2, MediaPipe, Kaldi, dlib
AI Foundations: LLM customization, fine-tuning, reinforcement learning, LangChain, LangGraph
Backend / Cloud: FastAPI, Pydantic, Django REST, Spring Boot, React, Docker, GCP, Git, Jira, Linux, AWS Lambda, S3, DynamoDB, Rekognition
Certifications: AWS Certified Solutions Architect – Associate; AWS Certified Cloud Practitioner - Associate AWS Certified Machine Learning Engineer - In progress

Relevant Coursework

- **Computational Science Foundations:** Computational Science Foundations for CMDA; Problem Solving in Computer Science; Discrete Mathematics; Multivariable Calculus; Differential Equations
- **Statistics & Modeling:** Probability & Distributions; Statistical Methods; Regression Analysis; Linear Algebra; Mathematical Modeling